Cryptography Lab 07: Classical Substitution, Vigenère Cryptanalysis & Transpositions

Authors: AJANA Chadi
Focus: Symmetric Algorithms, Frequency Attacks, Ciphertext Decryption, and Security Principles.

1. Part 1: Security Properties & Threat Modeling

We evaluated several threat scenarios to map them to core security properties:

1. **Scenario:** An attacker modifies the chemical configuration files of a medical manufacturing facility.
   - *Violated Property:* **Integrity**. Unauthorized alterations compromise the accuracy and reliability of the data.
2. **Scenario:** An attacker exfiltrates proprietary manufacturing formulas.
   - *Violated Property:* **Confidentiality**. Proprietary data was exposed to unauthorized entities.
3. **Scenario:** An attacker compromises cryptographic authentication keys.
   - *Violated Property:* **Authenticity / Identity Verification**. The system can no longer confirm whether transactions originate from a legitimate user or an impersonator.
4. **Scenario:** An attacker floods a manufacturing node, knocking the operations control console offline.
   - *Violated Property:* **Availability**. Authorized operators are blocked from accessing critical production services.
5. **Architectural Countermeasures:** Implement end-to-end data encryption for storage protection, strictly enforce role-based access control (RBAC), employ asymmetric digital signatures to verify data integrity, and maintain offline, redundant backups to protect against availability loss.

2. Part 2: Substitution Cipher Evaluation

Caesar Cipher Execution

- **Parameters:** Plaintext $P = \text{securite}$, Shift Key $K = 3$.
- **Algorithm:** For each character P_i, the shifted ciphertext value C_i is computed as:

$$C_i = (P_i + K) \pmod{26}$$

Mapping Table:

Plaintext P_i	s (18)	e (4)	c (2)	u (20)	r (17)	i (8)	t (19)	e (4)
$P_i + 3 \pmod{26}$	21	7	5	23	20	11	22	7
Ciphertext C_i	**V**	**H**	**F**	**X**	**U**	**L**	**W**	**H**

- **Resulting Ciphertext:** VHFXULWH

Monoalphabetic Custom Substitution

- **Given Alphabets:**
  - **Plain:** ABCDEFGHIJKLMNOPQRSTUVWXYZ
  - **Cipher:** NLEHXYJTGKZDARWMCQSBOFVIPU
- **Plaintext:** SECURITE

Substitution Process:

- $S \rightarrow \mathbf{S}$
- $E \rightarrow \mathbf{X}$
- $C \rightarrow \mathbf{E}$
- $U \rightarrow \mathbf{O}$
- $R \rightarrow \mathbf{M}$
- $I \rightarrow \mathbf{K}$
- $T \rightarrow \mathbf{B}$
- $E \rightarrow \mathbf{X}$
- **Resulting Ciphertext:** SXEOMKBX

3. Part 3: Vigenère Cipher Analysis & Cryptanalysis

Vigenère Decryption Implementation

- **Ciphertext (C):** vvqimsykttzbx
- **Key (K):** test (Extended key pattern: testtesttestt)
- **Mathematical Formula:**

$$P_i = (C_i - K_i) \pmod{26}$$

Calculation Sequence:

Cipher C_i :	v (21)	v (21)	q (16)	i (8)	m (12)	s (18)	y (24)	k (10)	t (19)	t (19)	z (25)	b (1)	x (23)
Key K_i :	t (19)	e (4)	s (18)	t (19)	t (19)	e (4)	s (18)	t (19)	t (19)	e (4)	s (18)	t (19)	t (19)
$C_i - K_i$:	2	17	−2	−11	−7	14	6	−9	0	15	7	−18	4
(mod 26) :	2	17	24	15	19	14	6	17	0	15	7	8	4
Plain P_i :	**c**	**r**	**y**	**p**	**t**	**o**	**g**	**r**	**a**	**p**	**h**	**i**	**e**

- **Recovered Plaintext:** cryptographie

Vigenère Cryptanalysis (Key Reconstruction)

- **Known Plaintext:** MASTER TELECOM (Normalized: MASTERTELECOM)
- **Captured Intercepted Ciphertext:** AKGDSBHOZOQYA
- **Reconstruction Formula:**

$$K_i = (C_i - P_i) \pmod{26}$$

Shift Verification:

- $M(12) \to A(0) \implies (0 - 12) = -12 \equiv 14 \to \mathbf{O}$
- $A(0) \to K(10) \implies (10 - 0) = 10 \to \mathbf{K}$
- $S(18) \to G(6) \implies (6 - 18) = -12 \equiv 14 \to \mathbf{O}$
- $T(19) \to D(3) \implies (3 - 19) = -16 \equiv 10 \to \mathbf{K}$
- $E(4) \to S(18) \implies (18 - 4) = 14 \to \mathbf{O}$
- $R(17) \to B(1) \implies (1 - 17) = -16 \equiv 10 \to \mathbf{K}$
- **Discovered Key Pattern:** OKOKOKOKOKOK
- **Extracted Cryptographic Key:** OK

4. Part 4: Transposition Cipher Operations

Columnar Transposition Configuration

- **Plaintext Input:** "seul celui qui voyage verra le long chemin qui mène à son pays"
- **Data Sanitization (Remove spaces, punctuation, and accents):** seulceluiquivoyageverralelongcheminquimeneasonpays
- **Operational Process:** The sanitized plaintext is loaded into a grid with a fixed column width (e.g., matching the length of a keyword like TELECOM). The ciphertext is then read column-by-column based on the alphabetical order of the key. This changes the structural positions of the letters without modifying the characters themselves.

Frequency Analysis Attack Vectors

- **Intercepted Text:** LHLZ HFQ BC HFFPZ WH YOUPFH MUPZH

- **Letter Distribution Counts:**
    - $H : 6$
    - $F : 4$
    - $P : 3$
    - $Z : 3$
    - $L : 2$
    - $U : 2$
- **Cryptanalysis Steps:**
    1. In standard English text, the letter **E** typically exhibits the highest natural frequency (~12.7%).
    2. Because **H** is the most frequent character in the ciphertext ($H = 6$), it is mapped to **E** as a starting hypothesis.
    3. The analyst iteratively tests substitutions for high-frequency letters, using pattern matching and common bigrams or trigrams to fully recover the plaintext.